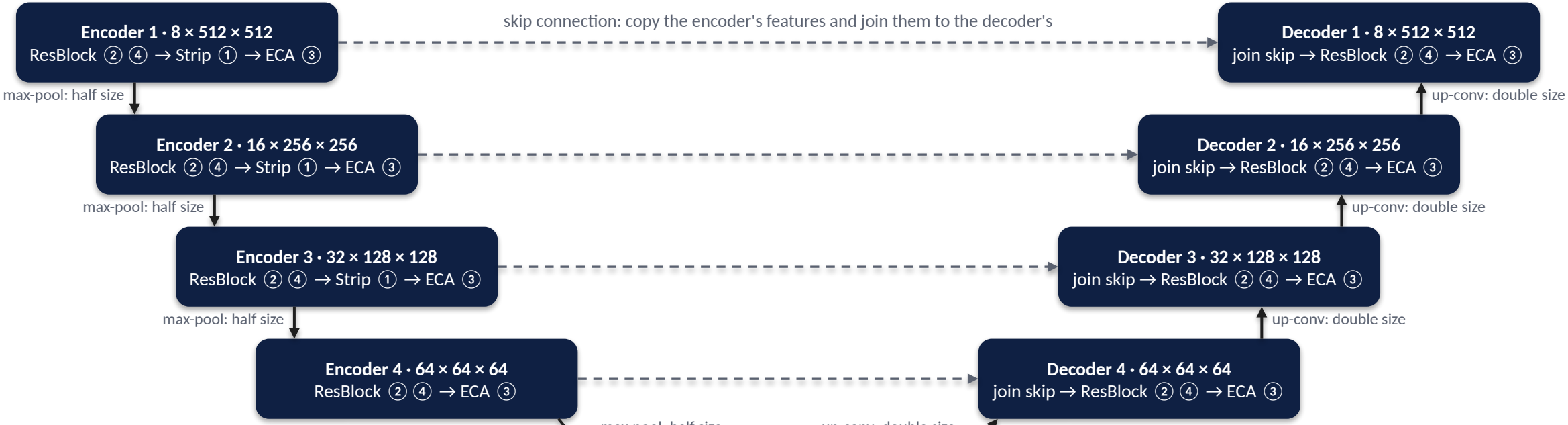
➢Our model: the same U, changed inside

▼ Input: spectrogram, 1 × 512 × 512 — no border trimming ⑦

1×1 conv → 2 raw scores → softmax ⑤ → RFI probability ▲



① **Strip convolutions** — 7/11/21-px windows (enc. 1–3, bottleneck)

② **Residual shortcut** — inside every ResBlock

③ **ECA attention** — after every encoder and decoder block

④ **GroupNorm** — in every ResBlock and strip (tf_unet: none)

⑤ **Raw scores** — no ReLU before softmax (tf_unet has one)

⑥ **Dice + cross-entropy loss** (tf_unet: cross-entropy + L2 penalty)

⑦ **Same padding** — every pixel gets a prediction

⑧ **No class weighting** — tf_unet has none either; our earlier version used it and scored lower

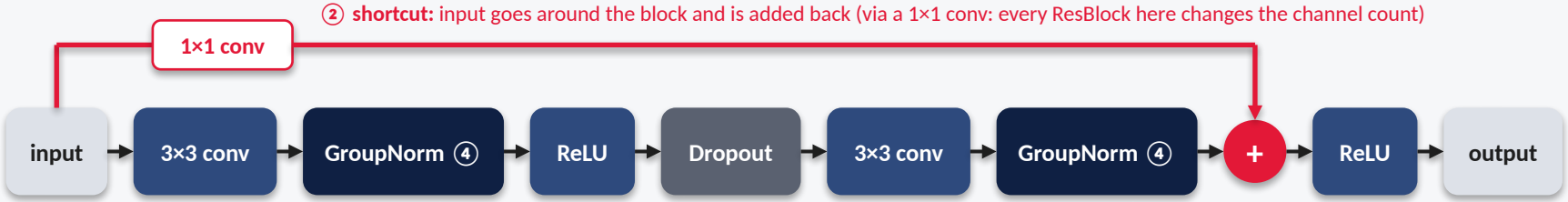
⑨ **Size**: shrinks 4 times, 8 → 128 channels, 593,842 parameters

Red = the three added modules ① – ③

Inside one level, step by step

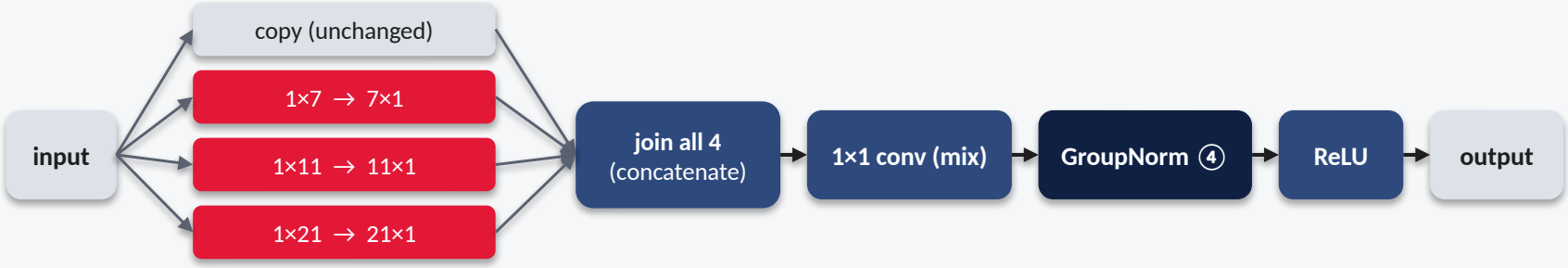
ResBlock ② ④

Two 3×3 convolutions, each tidied by GroupNorm; the input is added back at the end. Dropout works in training only.



Strip module ①

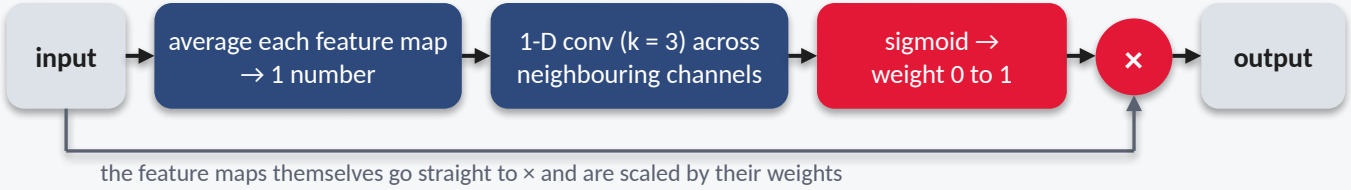
Looks at wider areas (7, 11, 21 px) cheaply: a thin filter along frequency, then one along time.



Each red path = 1×K along frequency, then K×1 along time (LOFAR axes), each channel on its own → together it covers a K×K square (measured)

ECA attention ③

Gives each feature map a weight between 0 and 1, then scales it.



the feature maps themselves go straight to × and are scaled by their weights